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EEG AND SLEEP

Electro Encephalography (EEG) 00:02:12

Records brain waves.

Represents summation of local potentials (Overall electrical activity of brain).

Characteristics of EEG waves :

Frequency :

- Number of waves/second.
- Frequency ∝ brain activity.

Amplitude :

- Height of wave.

Types :

Features	α wave	β wave	θ wave	δ wave	γ wave
Waveform					-
Significance	Wave of inattention (Relaxed state)	Wave of attention/ wakefulness	• memory. • Sleep.	Deep sleep	Focused attention
Frequency	8–12 Hz	13–30 Hz	4–7 Hz	1–3 Hz (Slowest)	> 60 Hz (max)
Source	Occipital & parietal cortex	Prefrontal cortex	Hippocampus (Short term → Long term memory)	-	-

α block : Replacement of α wave with β wave.

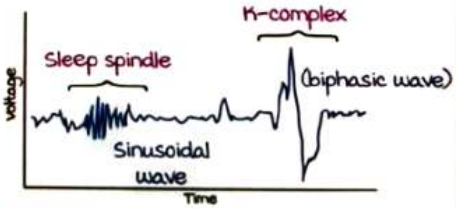
Sleep 00:17:06

TYPES

Non-rapid eye movement (NREM) sleep :

- Deep sleep.
- ↑ delta activity.

Stages :

Features	1	2	3	4
Frequency	4–7 Hz	8–12 Hz	-	-
Wave	Theta		Delta (Appears in stage 3, max. activity in stage 4)	
New nomenclature	N_1	N_2	N_3 (deep sleep)	

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Physiological changes :

1. RR : ↓.
2. BP : ↓.
3. Temperature : ↓↓.
4. Respiration : Slow & deep.

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Rapid eye movement (REM) sleep :

- EEG : β wave.
- AKA paradoxical sleep (EEG resembles wakefulness).

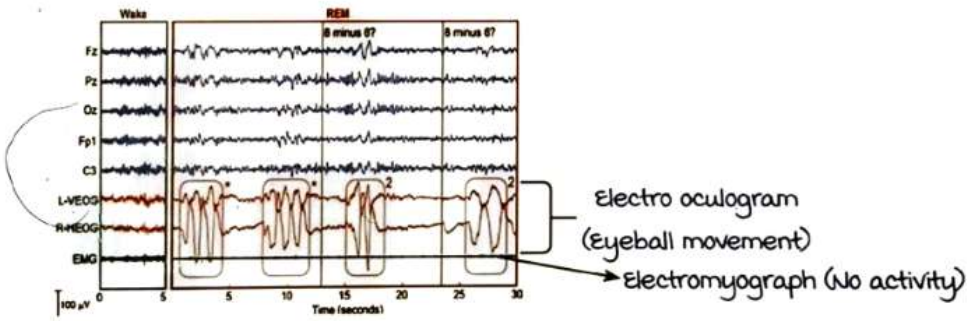
Eyeball movement :

Mechanism : Ponto-geniculo-occipital (PGO) spikes.

Pons
↓+
Lateral geniculate body activation of magnocellular pathway
↓+
Occipital lobe
↓
Eyeball movement

Physiological changes :

1. Muscle atonia → Prevents acting out of dreams.
   - Exceptions → Diaphragm : For respiration
   - Extra ocular muscles : For eyeball movement.
2. Genital organ (Penis/clitoris) enlargement.



HORMONAL CHANGES :

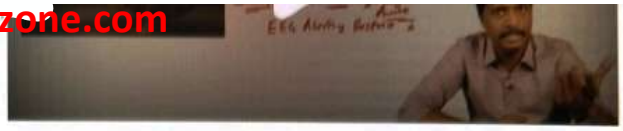
1. Growth hormone : ↑ in stages 3 & 4 of NREM sleep.
2. Prolactin : ↑ in REM sleep.
3. Luteinizing hormone : Nocturnal LH surge in puberty.
4. Melatonin : Sleep-inducing hormone secreted by pineal gland.

DREAMS :

Features	NREM sleep	REM sleep
Duration	Shorter	Longer (Early morning)
Type of dream	Conceptual	Emotional
Content of dream	Related to day to day activities	Unrelated to day to day activities

Feedback

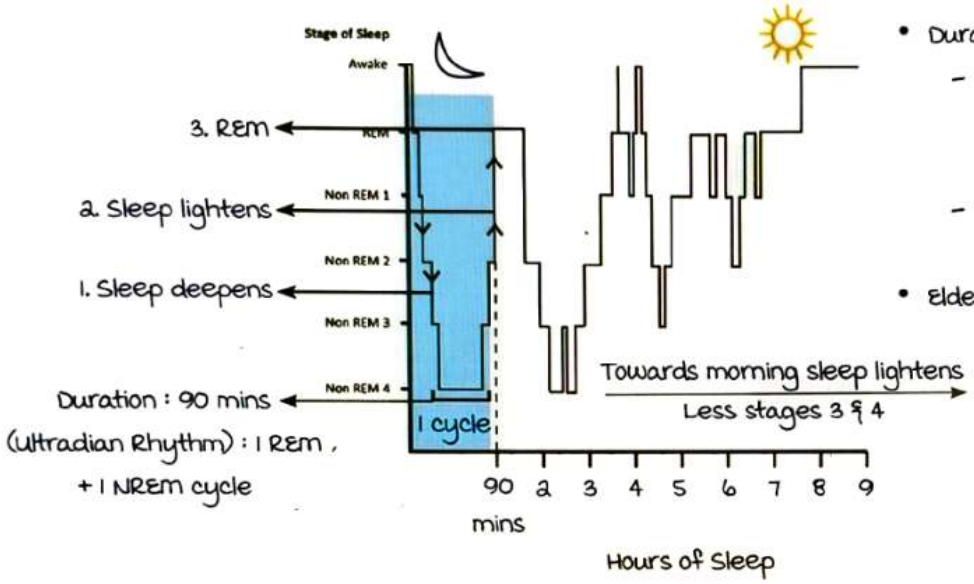
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SLEEP STUDY

Hypnogram :



- Duration :
  - Normal.
    - NREM : 75 % (Stage 2 : 45 %).
    - REM : 25 %.
  - Neonate :
    - REM : 50 %.
- Elders : Less deep sleep.

Regulation of sleep :

Areas promoting wakefulness :

Areas	Neurotransmitter
1. Reticular activating system (RAS)	Norepinephrine → Widespread brain activation
2. Raphe nuclei	Serotonin
3. Hypothalamus • Lateral	Orexin
• Posterior	Histamine

- EEG alerting response : High frequency stimulation of RAS → Awakens person.

Areas promoting sleep :

1. Hypothalamus : Ventrolateral pre optic nucleus (VLPON).
2. Cholinergic neurons → Acetylcholine (Facilitates REM sleep).
3. Pineal gland → Melatonin.

Mechanism :

NREM on neurons
↓
GABA
↓
Posterior hypothalamus
↓
Release inhibited ← ✕
↓
Histamine
↓
NREM sleep

REM on neurons
↓
Acetycholine
↓
REM sleep

Feedback



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PARASOMNIAS

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NREM parasomnias :

1. Sleep walking/somnambulism : In stages 3 & 4 (Deep sleep).
2. Sleep talking.
3. Bruxism/teeth grinding : In stage 2.
   - Seen in emotionally neglected children.
4. Nocturnal enuresis/bed wetting.
5. Night terrors/pavor nocturnus.

REM parasomnias :

1. Narcolepsy : ↑ day time sleepiness.
2. Nightmares : Terror dreams in REM.

Feedback